A

Micro Project report on

TRENDS IN IPR, FORMS OF IP, COPYRIGHT, TRADEMARKS, PATENTS, DESIGNS AND TRADE SECRETS

Submitted to the CMR Institute of Technology in partial fulfilment of the requirement of the

PROFESSIONAL PRACTICE, LAW & ETHICS LAB

IV-B. TECH I-SEMESTER

In

Computer Science and Engineering (Data Science)

Submitted By

R.M.N Sai Prabhas **23R01A67B6**

Under the Guidance of

Mrs K Swaroopa

(Assistant Professor, (H&S))



CMR INSTITUTE OF TECHNOLOGY

(UGC AUTONOMOUS)

(Approved by AICTE, Affiliated to JNTUH, Hyderabad)

Kandlakoya, Medchal Road, Hyderabad-501401.

2026-2027

CMR INSTITUTE OF TECHNOLOGY
(UGC AUTONOMOUS)
(Approved by AICTE, Affiliated to JNTUH, Hyderabad)
Kandlakoya, Medchal Road, Hyderabad-501401.
Computer Science and Engineering (Data Science)

CERTIFICATE

This is to certify that a Micro Project entitled with "**Trends in IPR, Forms of IP, Copyright, Trademarks, Patents, Designs and Trade Secrets**" being submitted by

R.M.N Sai Prabhas **23R01A67B6**

In partial fulfilment of the requirement for completion of the "PROFESSIONAL PRACTICE, LAW & ETHICS LAB" of IV-B.TECH I-semester of a bonafide work carried out under the guidance and supervision.

Signature of Faculty
Mrs K Swaroopa
(Assistant Professor,(H&S))

Signature of HOD
Dr.M.Ravi
(Head of Department,CSE(DS))

Signature of External Examiner

ACKNOWLEDGEMENT

I am extremely grateful to **Dr. M. Janga Reddy**, Director, **Dr. B. Satyanarayana**, Principal and **Dr. M. Ravi**, Head of Department, Computer Science and Engineering (Data Science), CMR Institute of Technology for their inspiration and valuable guidance during the entire duration.

I am extremely thankful to **Mrs K Swaroopa**, Assistant Professor, CSE(DS), CMR Institute of Technology for her constant guidance, encouragement and moral support throughout the project.

I express my thanks to all staff members and friends for all the help and coordination extended in bringing out this project successfully in time.

R.M.N Sai Prabhas **23R01A67B6**

INDEX

S.NO	CONTENTS	PAGE NO.
1	Abstract	1
2	Introduction	2
3	Objective	3
4	Need for the Study	4
5	Literature Review	5
6	Methodology	6
7	Detailed Study	7-11
8	Importance	12
9	Observations	13
10	Applications	14
11	Conclusion	15
12	References	16

ABSTRACT

Intellectual Property Rights (IPR) are legal rights given to the creators of original works and inventions, allowing them to control, use and earn from those creations for a fixed period. In a knowledge-based economy, ideas, brands, designs and data have become some of the most valuable assets of individuals, companies and nations, and IPR is the legal system that protects them.

This micro project studies the major forms of intellectual property – **Copyright, Trademarks, Patents, Designs and Trade Secrets** – and the trends that are changing IPR in India and across the world. For every form, the project explains what is protected, the Indian law that governs it, the conditions and duration of protection and the remedies against infringement. A comparative table brings the five forms together.

The study then examines recent trends, including the record growth in global patent and design filings reported by WIPO, the rapid rise of Indian filings, the debate on generative AI and copyright, the fight against online piracy and deepfakes, and the growing importance of trade secrets. These developments show that IPR is no longer a purely legal subject; it affects every engineer, developer and data scientist who writes code, trains models, names a product or shares work online.

The project concludes that awareness of IPR is both a legal necessity and an ethical responsibility: it helps professionals protect their own innovations, respect the work of others and contribute to a fair and innovative society. The report is educational in nature and is not legal advice.

The report is organised as follows: the objective, need and literature review set the background; the methodology explains how the study was carried out; the detailed study covers the forms of IP and emerging trends; and the remaining sections discuss the importance, observations and applications of IPR before concluding.

INTRODUCTION

Creativity and innovation drive economic and technological progress. Unlike land or machinery, creations of the human mind – inventions, books, music, software, logos and business know-how – are intangible and can be copied at almost no cost. If anyone could copy them freely, creators would have little reason to invest time and money in new work. Intellectual property law solves this problem by granting the creator a temporary exclusive right, while ensuring that knowledge eventually enters the public domain for the benefit of society.

Modern IP law developed in stages: the Statute of Monopolies (1623) and the Statute of Anne (1710) in England; the Paris Convention (1883) for industrial property; the Berne Convention (1886) for copyright; the World Intellectual Property Organization, WIPO (1967); and the WTO's TRIPS Agreement (1995), which sets minimum standards for all WTO members. In India, the main laws are the **Copyright Act, 1957**, the **Patents Act, 1970**, the **Trade Marks Act, 1999** and the **Designs Act, 2000**, supported by the National IPR Policy of 2016.

The five forms covered in this project are: **copyright**, which protects creative expression; **trademarks**, which protect brand identity; **patents**, which protect inventions; **designs**, which protect the visual appearance of products; and **trade secrets**, which protect confidential business information.

In recent years the field has been reshaped by digital technology, e-commerce and artificial intelligence. Online piracy, counterfeit goods, deepfakes and the use of copyrighted material to train AI models raise new questions. For computing students, source code, algorithms, datasets and trained models are all potential subject matter, so understanding IPR is a legal requirement as well as a moral obligation.

IPR is closely linked with professional ethics. Respecting another person's copyright, avoiding plagiarism, honouring confidentiality agreements and giving credit for ideas are ethical duties that the law also enforces. A professional who understands IPR can therefore innovate with confidence and avoid disputes.

OBJECTIVE

The main objective of this micro project is to understand the forms of intellectual property and the current trends in IPR, and to appreciate their importance in modern professional practice.

Specific Objectives:

1. To study the concept, purpose and international framework of Intellectual Property Rights and to understand how they promote creativity, innovation and ethical practice in society.
2. To explain Copyright, Trademarks, Patents, Designs and Trade Secrets and the Indian laws that govern each of them, including what each form protects and for how long.
3. To compare the five forms on scope, registration, duration and conditions of protection so that the right form can be chosen for a given creation.
4. To analyse current trends such as record global filings, the growth of Indian filings, AI and copyright, digital enforcement and personality rights.
5. To understand the ethical responsibilities of engineers and professionals in using and protecting intellectual property.
6. To highlight the importance of IPR awareness for students, startups and professionals in a competitive, knowledge-based economy.

NEED FOR THE STUDY

Intellectual property is now one of the most valuable assets of modern organisations. From startups and research institutions to large industries, every enterprise depends on its patents, brands, software and know-how to stay competitive. A clear understanding of IPR is therefore essential for anyone who will create, manage or use such assets.

Awareness, however, remains limited among students and young professionals. This often leads to plagiarism, use of pirated software, breach of open-source licence conditions, careless disclosure of inventions before filing a patent, or the adoption of product names that infringe existing trademarks. Such mistakes can cause legal disputes and damage a professional's reputation.

The subject is also changing quickly. Generative AI, online platforms and global e-commerce have created questions that older laws did not anticipate, such as who owns AI-generated output and how counterfeit goods or pirated films can be stopped on the internet. At the same time, India's growing number of patent and design filings shows that more Indian innovators now need to protect their work.

For these reasons, a consolidated study of the forms of IP and the latest trends is necessary. It builds legal awareness, supports ethical professional conduct and prepares future engineers to protect their own innovations while respecting the rights of others.

The study is also relevant to career preparation. Engineers regularly work with patented technologies, licensed software and confidential client data, and many will go on to join or found startups where IP decisions – what to patent, what to keep secret, which name to register – affect the survival of the business.

LITERATURE REVIEW

The foundations of international IP law are the Paris Convention (1883), the Berne Convention (1886) and the TRIPS Agreement (1995). WIPO's *Intellectual Property Handbook* explains the concepts and policy of each form of IP, while TRIPS, administered by the World Trade Organization, requires member countries to provide minimum standards of protection and enforcement.

In India, the primary sources are the Copyright Act, 1957, the Patents Act, 1970, the Trade Marks Act, 1999 and the Designs Act, 2000, together with their rules. Standard textbooks by P. Narayanan and V. K. Ahuja discuss these Acts in detail. Judicial decisions such as *Novartis v. Union of India* (2013) on Section 3(d) and *Natco v. Bayer* (2012) on compulsory licensing show how courts balance private rights with public interest.

Policy literature includes the National IPR Policy (2016) of the Government of India, the 289th Report of the Law Commission of India (2024) on trade secrets and economic espionage, and the DPIIT committee's working paper on generative AI and copyright (December 2025). Statistical information on global trends is published each year in WIPO's *World Intellectual Property Indicators*, the 2025 edition of which is used in this study.

For international filings, the Patent Cooperation Treaty (1970) and the Madrid System for trademarks are the main routes, and India has been a member of the PCT since 1998 and of the Madrid Protocol since 2013. These instruments reduce the cost and effort of seeking protection in several countries.

The literature shows that strong and balanced IP protection encourages innovation, investment and technology transfer, but it also reveals open issues: enforcement in the digital environment, the legal status of AI-generated works, the absence of a dedicated trade secret law in India and limited IP awareness among students and small businesses. This project brings these sources together in a single, student-friendly study.

METHODOLOGY

This micro project follows a qualitative and descriptive research approach based entirely on secondary sources. The steps followed were:

1. **Data collection:** information was gathered from the Indian IP Acts and rules, WIPO and WTO publications, WIPO's World Intellectual Property Indicators 2025, DPIIT and Law Commission documents, standard textbooks and reported case decisions.
2. **Concept identification:** the literature was reviewed to identify the key concepts, the five forms of IP and the international treaties that support them.
3. **Legal analysis:** each form was analysed under a common set of headings – subject matter, governing law, conditions for protection, duration, registration and remedies.
4. **Comparative tabulation:** the five forms were compared in a single table to show their similarities and differences.
5. **Trend identification:** current trends were identified from WIPO statistics, recent policy proposals and court decisions.
6. **Documentation:** the findings were organised into sections and presented in a structured report.

Scope and limitation: no field survey or primary data collection was carried out, and the report is intended for educational purposes only. Legal provisions and statistics may change, so the original Acts and official publications should be consulted for authoritative information.

DETAILED STUDY

1. Overview of Intellectual Property Rights (IPR)

Intellectual Property Rights are legal rights that protect creations of the human mind, such as inventions, literary and artistic works, designs, symbols, names and confidential business information. The creator receives an exclusive right to use and commercially exploit the creation for a limited period, and in return society benefits from the disclosure and eventual free use of that knowledge. IPR has an economic side (the right to earn from the work) and a moral side (the right to be recognised as its author). These rights are generally territorial: protection granted in one country does not automatically extend to another.

At the international level, protection is coordinated through treaties administered by WIPO and the WTO. The **Paris Convention** (industrial property) and the **Berne Convention** (copyright) establish national treatment; the **Patent Cooperation Treaty (PCT)** and the **Madrid System** allow a single international application for patents and trademarks; and **TRIPS** sets minimum standards of protection and enforcement for all WTO members. India is a party to these arrangements and has aligned its laws with them.

2. Forms of Intellectual Property

IP is commonly divided into **copyright and related rights**, which protect creative expression, and **industrial property**, which protects inventions, brands and the appearance of products in trade. Confidential commercial information forms a third category, recognised in Article 39 of TRIPS as "undisclosed information". Besides the five forms studied here, India has special laws for Geographical Indications (1999), semiconductor layout-designs (2000) and plant varieties (2001).

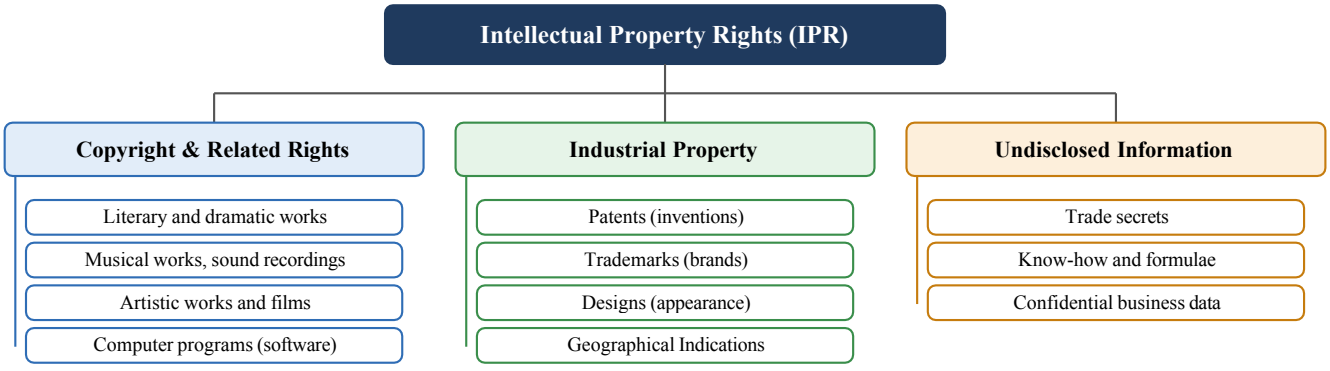


Fig. 1: Classification of Intellectual Property Rights

a) Copyright

Copyright protects the *expression* of ideas, not the ideas themselves. Under the Copyright Act, 1957 it subsists in original literary, dramatic, musical and artistic works, cinematograph films and sound recordings. Computer programs are protected as literary works, so source code is covered. Protection arises automatically on creation; registration with the Copyright Office is optional but is useful evidence of ownership.

- **Rights:** Section 14 gives economic rights to reproduce, publish, perform, translate, adapt and communicate the work to the public; Section 57 gives the author moral rights of paternity and integrity, which survive assignment of the economic rights.
- **Term:** the life of the author plus 60 years for literary, dramatic, musical and artistic works; 60 years from publication for films, sound recordings and photographs.
- **Exceptions:** Section 52 permits fair dealing for private study, research, criticism, review and reporting, and certain educational uses.
- **Infringement:** copying a substantial part without permission is infringement (Section 51); remedies include injunctions, damages and criminal penalties. Open-source licences such as MIT and GPL are copyright licences that set conditions for reusing code.

b) Trademark

A trademark is a mark capable of being represented graphically and of distinguishing the goods or services of one person from those of others – a word, logo, slogan, shape of goods, packaging, colour combination or sound. It is governed by the **Trade Marks Act, 1999** and the Trade Marks Rules, 2017, administered by the Trade Marks Registry under the Controller General of Patents, Designs and Trade Marks (CGPDTM).

- **Procedure:** search, application in one or more of the 45 Nice classes (34 for goods, 11 for services), examination, publication in the Trade Marks Journal, opposition and registration.
- **Distinctiveness:** arbitrary or fanciful marks (such as “Apple” for computers) are strongest; generic or descriptive marks are refused under Section 9, and marks confusingly similar to earlier ones under Section 11.
- **Term:** 10 years, renewable indefinitely for further 10-year periods.
- **Enforcement:** Section 29 covers infringement of registered marks; unregistered marks are protected by the common-law action of *passing off*. India is a member of the Madrid Protocol (2013) for international registration.

c) Patent

A patent is an exclusive right granted for an invention – a product or process – that is **new, involves an inventive step and is capable of industrial application** (Section 2(1)(j)). In exchange, the inventor must fully disclose the invention to the public. Patents are governed by the **Patents Act, 1970** (amended in 1999, 2002 and 2005) and the Patents Rules, 2003.

- **Procedure:** filing (a provisional specification secures an early priority date), publication after 18 months, request for examination, examination report, opposition and grant.
- **Term and rights:** 20 years from the date of filing (Section 53). The patentee can stop others from making, using, selling or importing the invention in India. Patents are territorial; foreign protection is sought through the PCT, which India joined in 1998.
- **Non-patentable matter:** Section 3 excludes, among others, mere discoveries, new forms of known substances without enhanced efficacy (Section 3(d), applied in *Novartis v. Union of India*, 2013) and "a mathematical or business method or a computer programme per se or algorithms" (Section 3(k)). Software-related inventions can be patented only if they show a technical effect, as explained in the CRI Guidelines.
- **Public interest:** compulsory licensing under Section 84 balances private rights with access, as in *Natco v. Bayer* (2012) for a cancer drug.

d) Industrial Design

Under the **Designs Act, 2000**, a design means the features of shape, configuration, pattern, ornament or composition of lines or colours applied to any article, in two or three dimensions, by an industrial process, which appeal to and are judged solely by the eye. It protects how a product *looks*, not how it works – for example, the contour of a mobile phone or a bottle. A design must be new and original, must not have been published anywhere before the filing date and must not be contrary to public order or morality.

A registered design is protected for **10 years, extendable once by 5 years** (maximum 15 years), and articles are classified under the Locarno Classification. Under Section 15 of the Copyright Act, copyright in a registrable design ceases once the article has been reproduced more than 50 times by an industrial process, so a manufacturer must register the design to keep protection.

e) Trade Secret

A trade secret is confidential business information that has commercial value because it is secret and is subject to reasonable steps to keep it secret – for example, manufacturing processes, formulae, customer lists, algorithms and unpublished source code. **India has no dedicated trade secret statute.** Protection comes from contract law (NDAs and employment agreements), the equitable doctrine of breach of confidence, and related provisions of the Copyright Act and the Information Technology Act, 2000. Post-employment non-compete clauses are generally void under Section 27 of the Indian Contract Act, 1872, but confidentiality obligations remain enforceable. In 2024 the Law Commission of India recommended a dedicated trade secrets law.

Unlike a patent, a trade secret needs **no registration and no disclosure** and can last indefinitely while secrecy is maintained. However, it gives no protection against independent discovery or lawful reverse engineering, and once leaked it is lost for good. Good practice includes NDAs, need-to-know access and marking documents as confidential. Well-known examples are the Coca-Cola formula and the Google search algorithm.

Comparative analysis of the five forms

Parameter	Copyright	Trademark	Patent	Design	Trade Secret
Protects	Original literary, artistic, musical works, software, films	Distinctive marks that identify goods or services	New, inventive and useful inventions	Visual appearance of an article	Confidential information of commercial value
Indian law	Copyright Act, 1957	Trade Marks Act, 1999	Patents Act, 1970	Designs Act, 2000	Contract law and equity (no dedicated Act)
Registration	Optional; arises automatically	Optional but advisable	Mandatory	Mandatory	Not possible
Duration	Life + 60 years (usual works)	10 years, renewable indefinitely	20 years from filing	10 years + 5 years extension	Indefinite while secrecy lasts
Key condition	Originality	Distinctiveness	Novelty, inventive step, industrial use	Novelty and originality	Secrecy and reasonable protective steps
Authority	Copyright Office	Trade Marks Registry	Patent Office	Design Office	Civil courts

Table 1: Comparison of the five forms of IP in India

3. Emerging Trends in IPR

a) Record global filings. WIPO's World Intellectual Property Indicators 2025 report about 3.7 million patent applications in 2024 (up 4.9%, the fifth straight year of growth) and 1.6 million design applications (up 2.2%), while trademark filings of about 15.2 million classes were essentially flat.

b) India's rise. India ranked sixth in the world for patent filings in 2024, and its design filings grew by about 45%. Growth is driven by resident applicants, supported by the National IPR Policy 2016, CIPAM, fast-track examination and reduced fees for startups.

c) Generative AI and copyright. Using copyrighted works to train AI models, and ownership of AI-generated output, are the most debated issues; the UK and USA have so far required a human inventor or author. In December 2025 a DPIIT committee published Part 1 of its working paper proposing a *hybrid model* – a mandatory blanket licence for AI training with a statutory right to remuneration for creators – while *ANI v. OpenAI* is before the Delhi High Court.

d) Digital enforcement and personality rights. Courts grant "dynamic injunctions" against mirror piracy websites (*UTV Software v. 1337x.to*, Delhi High Court, 2019) and protect the name, image and voice of public figures against deepfakes (for example, *Anil Kapoor*, 2023).

e) Strategic technologies and IP as an asset. Green-technology patents, standard-essential patents for 5G with FRAND licensing disputes, and reliance on trade secrets for AI models and datasets show how firms combine different forms of IP, increasingly treating IP as a financial asset through licensing and valuation.

f) Awareness and startups. Government schemes such as Startup India provide IP facilitation and reduced fees, and universities are setting up IPR cells, reflecting a push to build IP awareness among students and young innovators.

IMPORTANCE

Encourages innovation. Assured of legal protection, individuals and organisations invest in research, writing, design and development, which leads to continuous technological and cultural progress.

Gives legal ownership and remedies. Creators can stop unauthorised copying and claim compensation through injunctions, damages and, in many cases, criminal action.

Supports economic growth and trade. IP-intensive industries such as software, pharmaceuticals and entertainment contribute to income and employment, and strong IP systems help attract foreign investment and technology transfer. TRIPS ensures that protection is available across borders.

Builds brand value and consumer trust. Trademarks and designs let businesses distinguish their products, build reputation and protect consumers from counterfeits.

Helps startups. Registered IP adds credibility, supports licensing and helps startups attract investors.

Promotes professional ethics. Work created in the course of employment under a contract of service generally belongs to the employer (Section 17, Copyright Act). Professionals must therefore honour NDAs, follow open-source licence terms, give proper attribution and avoid plagiarism and pirated software.

Balances private and public interest. Time limits, fair dealing and compulsory licensing ensure that exclusive rights do not prevent society from eventually benefiting from knowledge.

Overall, IPR gives creators a fair return on their effort, gives businesses a secure basis for investment and ensures that society ultimately gains from the new knowledge that is created.

OBSERVATIONS

The study led to the following observations:

1. **Different forms, different protection:** each form protects a different aspect of a creation, and a single product – such as a smartphone – may involve patents, designs, trademarks, copyright in software and trade secrets together.
2. **Registration varies:** patents, designs and trademarks are strengthened by registration, copyright arises automatically, and trade secrets cannot be registered at all.
3. **Duration varies widely:** from 20 years for patents to indefinite protection for trademarks (on renewal) and trade secrets (while secrecy lasts).
4. **India's IP activity is growing:** WIPO data show Indian patent and design filings rising strongly, supported by government policy.
5. **Technology is moving faster than law:** AI-generated works and AI training data raise questions that existing statutes do not fully answer.
6. **Trade secrets are gaining importance** as firms protect data and models, yet India still lacks a dedicated statute.
7. **Enforcement is moving online:** courts increasingly use dynamic injunctions and takedown orders to deal with piracy and deepfakes.
8. **Awareness gap:** students and young professionals often lack knowledge of IP procedures and of their ethical duties under IP law.

APPLICATIONS

The five forms of IP are used in commercial and non-commercial sectors alike, often in combination. Some of the main areas of application are given below.

1. Technology and engineering: patents protect new technical solutions in electronics, telecommunications and energy, helping firms recover research costs.

2. Software and IT: copyright protects source code, patents protect technical inventions that meet Indian law, and trade secrets protect algorithms; trademarks identify software products.

3. Pharmaceuticals and healthcare: product and process patents protect new drugs and medical devices, with safeguards such as Section 3(d) and compulsory licensing.

4. Business, brands and e-commerce: trademarks, logos and packaging build brand identity, while anti-counterfeiting actions protect marketplaces and consumers.

5. Media and entertainment: copyright and licensing govern films, music, books and OTT content, and personality rights protect public figures.

6. Fashion and industrial design: design registration prevents imitation of the appearance of products such as clothing, furniture and consumer devices.

7. Academia and research: IPR cells and technology transfer offices help students and faculty file patents and license their inventions.

8. Startups: IP adds credibility, attracts investors and protects against idea theft, supported by Startup India IPR facilitation.

CONCLUSION

Intellectual property is the legal backbone of the knowledge economy. Copyright protects creative expression, trademarks protect brand identity, patents protect technical inventions, designs protect visual appearance and trade secrets protect valuable confidential information. Each form differs in subject matter, conditions, duration and registration requirements, and the right choice – or combination – depends on the nature of the creation.

The study of current trends shows that IPR is evolving rapidly. Record global filings, the rise of Indian innovators, the debate on generative AI and copyright, stronger online enforcement and the growing role of trade secrets will keep reshaping the field. Future developments are likely to include clearer laws on AI training data and AI-generated works, a dedicated trade secret statute in India, faster digital IP offices and better cross-border enforcement.

For engineers, developers and data scientists, a sound understanding of IPR is both a legal necessity and an ethical responsibility. It protects their own innovations, ensures respect for the work of others and supports a fair, innovation-driven society.

Practical recommendations drawn from the study:

- Keep dated records of original work and use proper copyright notices.
- Search existing trademarks before adopting a product or company name.
- Keep an invention confidential until a patent application is filed.
- Register a design before mass production begins.
- Protect trade secrets with NDAs and access controls.
- Read and follow the licence terms of every library, dataset and model you use.

REFERENCES

1. Government of India, *The Copyright Act, 1957* (Act 14 of 1957), as amended in 2012.
2. Government of India, *The Patents Act, 1970* (Act 39 of 1970), as amended; The Patents Rules, 2003.
3. Government of India, *The Trade Marks Act, 1999* (Act 47 of 1999); The Trade Marks Rules, 2017.
4. Government of India, *The Designs Act, 2000* (Act 16 of 2000); The Designs Rules, 2001.
5. World Trade Organization, *Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS)*, 1994.
6. WIPO, *Paris Convention for the Protection of Industrial Property* (1883) and *Berne Convention for the Protection of Literary and Artistic Works* (1886).
7. WIPO, *World Intellectual Property Indicators 2025*, Geneva, 2025.
8. WIPO, *WIPO Intellectual Property Handbook: Policy, Law and Use*, WIPO Publication No. 489.
9. Department for Promotion of Industry and Internal Trade (DPIIT), *National Intellectual Property Rights Policy*, 2016.
10. DPIIT Committee, *Working Paper on Generative AI and Copyright (Part 1)*, December 2025.
11. Law Commission of India, *Report No. 289: Trade Secrets and Economic Espionage*, 2024.
12. Office of the Controller General of Patents, Designs and Trade Marks, *Intellectual Property India*, www.ipindia.gov.in.
13. Narayanan, P., *Intellectual Property Law*, Eastern Law House.
14. Ahuja, V. K., *Law Relating to Intellectual Property Rights*, LexisNexis.